Exercise Set 2 — Defense Sheet

Computational Finance (WI4151), TU Delft — Prof. Fang Fang

Contents

This set has two problems, each in its own file: - **Exercise 1 — Least-Squares Monte Carlo (LSM / Longstaff–Schwartz):** `exercise1_def (2).py` - **Exercise 2 — Finite Difference (FTCS) for European and American puts:** `exercise2_def (2).py`

Line numbers below refer to those two files. Where a "why" requires my own reasoning, it is left as a $TODO(Bruno)$ for me to answer out loud.

---

1 The task (in my own words)

Exercise 1 — LSM (Longstaff–Schwartz, 2001). - Part (i): reproduce the textbook 8-path numerical example from the paper. Strike $K = 1.10$, $r = 6$, three yearly exercise dates ($dt = 1$, $T = 3$). Print the cash-flow matrix at each time, the in-the-money regression tables, the fitted $E[Y|X] = a0 + a1X + a2X^2$, the optimal early-exercise decisions, the stopping rule, the option cash-flow matrix, and finally the American and European put prices to 5 decimals. - Part (ii): replace the 8 hard-coded paths with $N = 100,000$ simulated GBM paths (50k + 50k antithetic), $K = 40$, $M = 50$ exercise dates per year, and replicate the "red box" columns of Table 1 of the paper (Simulated American, s.e., closed-form European, early-exercise value) for the 20 $(S0, sigma, T)$ combinations. Here the basis is the first three weighted Laguerre polynomials evaluated at `x = S/K`.

Exercise 2 — Finite Difference (FTCS). - Part (i): adapt a provided `ftcs_bs_put.py` to price

European puts via an explicit (forward-time, central-space) scheme in log-price space $m = logS$, for the same 20 parameter combinations, and compare against closed-form Black–Scholes. - Part (ii): turn the European FTCS solver into an American one by adding the early-exercise (free-boundary) projection $U = max(U, intrinsic)$ after each time step. Compare FD-American against the LSM-American of Exercise 1 as a correctness check. - Part (iii): assemble the full "outside the red box" columns of Table 1 (FD American, closed-form European, FD early-exercise value, and the FD-minus-LSM difference in early-exercise value).

TODO (Bruno): In one sentence, why does the American put price exceed the European put price here, and why is that *not* true for an American call on a non-dividend stock?

---

2 My approach and why

Exercise 1 — LSM. - Backward induction over exercise dates. At each date, regress discounted realized future cash flows Y on a basis in the current spot X, but **only over in-the-money paths**, then compare immediate exercise against the fitted continuation value. - Part (i) uses a degree-2 monomial basis $1, X, X^2$ via an sklearn pipeline (`exercise1_def (2).py:107`). - Part (ii) uses the first three weighted Laguerre polynomials in `x = S/K` (`exercise1_def (2).py:185-191`). - Variance reduction by antithetic paths (`exercise1_def (2).py:177-178`).

TODO (Bruno): Why regress only on in-the-money paths rather than all paths? (Out-of-the-money continuation values are irrelevant to the exercise decision — but say *why* in your own words and tie it to the paper.)

TODO (Bruno): Why monomials $1, X, X^2$ in Part (i) but weighted Laguerre in Part (ii)? Was this just "match the paper," or numerical conditioning? Be ready to defend both.

TODO (Bruno): Antithetic variates: why does pairing Z with $-Z$ reduce variance for this payoff, and what property of the estimator makes the variance reduction work? State the covariance argument.

Exercise 2 — FTCS. - Work in log-price $m = logS$ so the Black–Scholes PDE has constant coefficients, build the explicit evolution matrix F once per parameter set, and march forward in transformed time (time-to-maturity). - American: project onto the payoff after each step (`exercise2_def (2).py:105`).

TODO (Bruno): Why transform to log-price $m = logS$ before discretizing instead of working directly in S? (Constant coefficients / uniform grid — but explain the consequence for the matrices $T1$, $T2$.)

TODO (Bruno): FTCS is *explicit*. Why is that an acceptable choice here, and what stability condition are you implicitly relying on with $Nt = 40000 * T$? (See "hardest question" below — this is the danger zone.)

---

3 Key code sections (file + what it does + why I wrote it that way)

3.1 Exercise 1 — `exercise1_def (2).py`

- `:11-24` — Part (i) parameters and the 8 hard-coded paths from the paper. C (cash-flow matrix) initialized to zeros at `:27`, shape $(8, 3)$.

- `:32-86` — three printing helpers (`print_formatted_table`, `print_regression_table`, $optimal_table$) that produce the per-time tables shown in the report. Pure presentation, no pricing logic.
- `:89` $LSQM(S, C, dt, T, K, r)$ — the Part (i) LSM engine.
  - `:91` terminal payoff $C[:, T-1] = max(K - S[:, T], 0)$.
  - `:93` $C_european$ keeps a copy of the terminal payoff for the European price later.
  - `:96` backward loop $i = T - 2...0$.
  - `:98` regressor $X = S[:, i+1]$ masked by $< K$ (so OTM entries are zeroed before the ITM filter), reshaped to 2D.
  - `:99-101` builds Y = sum of *all* future cash flows discounted back to time $i+1$ using $discount_factors = exp(-r * dt * (u - i))$ with $u = arange(i+1, ncols)$.
  - `:102-104` ITM mask $S[:, i+1] < K$; restrict X, Y to ITM.
  - `:107-108` degree-2 polynomial + linear regression pipeline, fit on ITM data.
  - `:111-112` extracts intercept $a0$ and coefficients $a1, a2$.
  - `:117-120` continuation = model prediction on ITM paths; exercise = $K - S$ on ITM paths.
  - `:122-127` exercise rule: if $exercise > continuation$, set $C[j, i]$ to exercise value and **zero out all later cash flows** $C[j, i+1:] = 0$; else $C[j, i] = 0$.
  - `:131` stopping matrix from $C! = 0$.
  - `:147-152` American price: discount each path's cash flows by $exp(-r * dt * t)$ for $t = 1..T$, sum per path, average.
  - `:156` European price: $mean(C_european * exp(-r * dt * T))$.
- `:167-173` — Part (ii) parameters ($K = 40$, $M = 50$, $N = 100000$).
- `:175-183` $simulate_paths$ — GBM exact-step simulation; `:177` draws `N//2` normals, `:178` stacks $-Z$ for antithetics; `:181-182` exact log-Euler update $S[:, i+1] = S[:, i] * exp((r - 0.5sigma^2)dt + sigmasqrt(dt)Z)$.
- `:185-191` $laguerre_basis$ — `x=S/K`; `L0=exp(-x/2)`, `L1=exp(-x/2)(1-x)`, `L2=exp(-x/2)(1-2x+x^2/2)`; stacked column-wise.
- `:193-196` `black_scholes_put` — closed-form European put.
- `:198-235` $LSM_pricing$ — Part (ii) LSM engine; same backward structure as $LSQM$ but basis is $laguerre_basis$ (`:212`) and $LinearRegression()$ with default intercept (`:215`). Returns per-path discounted cash flows.
- `:237-267` — loop over the 20 $(S0, sigma, T)$ combos: $M_total = M * T$, `dt = 1/M`, simulate, price, compute $american = mean(cashflow)$, `european = black_scholes_put(...)`, $ee = |american - european|$ (`:257`), `se = std(cashflow)/sqrt(N)` (`:258`).

TODO (Bruno): `:91` and `:203` index $S[:, T]$ / $S[:, M_total]$ for the terminal payoff while the cash-flow matrix C has only T (resp. M_total) columns indexed $0..T-1$. Explain the off-by-one convention: S has one more column (the $t = 0$ spot) than C. Be ready to point at exactly which column of S is "today" and which is maturity.

TODO (Bruno): `:257` uses `ee = np.abs(american - european)` (absolute value) in the Part (ii) block of `exercise1_def`, but the copy in `exercise2_def (2).py:223` uses the *signed* `ee = american_value - european_value`. Which one matches the paper's "early exercise value" column, and is the absolute value ever wrong (can LSM-American dip below European by noise)?

TODO (Bruno): Standard error `se = std(cashflow)/sqrt(N)` at `:258`. With antithetic sampling the N draws are **not independent** (each path is paired with its antithetic). Is dividing by $sqrt(N)$ the correct s.e., or does it understate/overstate it? Defend your formula.

3.2 Exercise 2 — `exercise2_def (2).py`

- `:5-14` — parameters $K = 40$, $r = 0.06$, and the 20 $(S0, sigma, T)$ combos.
- `:18-64` — **Part (i) European FTCS** loop:
  - `:24-26` closed-form BS put for reference.
  - `:29` domain length $L = 2 * log(K) + 2$; `:30` $Nx = 1000$; `:31` $Nt = 40000 * T$; `:32-33` steps `h = L/Nx`, `k = T/Nt`.
  - `:35-36` central-difference matrices: $T1$ (first derivative, `+1/-1` off-diagonals), $T2$ (second derivative, -2 on diagonal, 1 off-diagonals), size $(Nx - 1)$.
  - `:38-40` explicit evolution matrix `F = (1 - r k) I + 0.5 k sigma^2 / h^2 * T2 + k (r - 0.5 sigma^2)/(2h) * T1`.
  - `:42-43` grid $mvec$, centered and shifted by $+log(K)$.
  - `:45-46` initial condition $U[:, 0] = max(K - exp(mvec), 0)$ (payoff as the time-0 condition in time-to-maturity).
  - `:48-52` forward march; `:51` boundary correction term $p[0]$ injected at the low-m (deep ITM) boundary node, scaled by $K * exp(-r * time2mat)$.
  - `:54` `np.interp` to read off the price at $log(S0)$.
- `:66-118` — **Part (ii) American FTCS** loop: identical to Part (i) except $intrinsic = max(K - exp(mvec), 0)$ (`:95`) and the single added projection line `:105` `U[:,i+1] = np.maximum(U[:,i+1], intrinsic)`. `ee = ftcs_american - bs_put` (`:108`).
- `:120-204` — re-imports and re-runs the **Exercise 1 Part (ii) LSM** code (paths, Laguerre basis, BS put, $LSM_pricing$) to regenerate $results_lsmc$ for the cross-check.
- `:235-248` — correctness test table: $|FDAmerican - LSMAmerican|$ per row.
- `:250-261` — full "outside the red box" table including the signed $FDEE - LSMEE$ difference.

TODO (Bruno): `:29` $L = 2 * log(K) + 2$. Why this width, and why center the grid at $log(K)$ (`:43`)? What goes wrong at short maturity / high vol if the domain is too narrow — and what is the analogy with the COS truncation-range $[a, b]$ issue I hit in Assignment 2?

TODO (Bruno): `:51` the boundary term $p[0]$. Explain in words what Dirichlet boundary condition this enforces at the deep-in-the-money edge, why it carries the factor $K * exp(-r * time2mat)$, and what happens at the *other* (high-S) boundary where no correction is added.

TODO (Bruno): `:105` is described in the report as "the single line that changes." Explain why projecting $U = max(U, intrinsic)$ after each explicit step is a valid (operator-splitting / explicit-penalty) treatment of the American free boundary, and what its accuracy cost is versus a proper LCP/PSOR solve.

---

4 Design decisions I must justify

1. **ITM-only regression** (`:103-104`, `:211-213`). Factual: the mask is $S[:, i + 1] < K$. **TODO (Bruno):** justify why this is the Longstaff–Schwartz prescription and not a shortcut.
2. **Basis choice:** monomials degree 2 in Part (i) (`:107`) vs weighted Laguerre in Part (ii) (`:185-191`). Factual: the report says normalization `x=S/K` avoids underflow in `exp(-x/2)` for `S in [36,44]`. **TODO (Bruno):** confirm you can derive the underflow argument and say why monomials were fine for the small $K = 1.10$ example.
3. **Antithetic variates** (`:177-178`). **TODO (Bruno):** justify and quantify expected variance reduction.
4. **Exact GBM step** (`:182`) rather than Euler. **TODO (Bruno):** why is the log update exact for

GBM, and does discretization bias still enter the American price through the discrete exercise dates regardless?

5. $M = 50$ **exercise dates/year as a proxy for continuous American exercise** (`:172`, `:251`). **TODO (Bruno):** why is a Bermudan with 50/100 dates a good stand-in for the true American, and which direction is the bias?
6. **Explicit FTCS with** $Nt = 40000 * T$ (`exercise2_def (2).py:31`). **TODO (Bruno):** show that this choice keeps the scheme inside the explicit stability region (compute the CFL-type bound; see hardest question).
7. **`np.interp` to extract the price at** $log(S0)$ (`:54`, `:107`). Factual: linear interpolation between grid nodes. **TODO (Bruno):** is linear interpolation accurate enough given h, or should it be higher order?

---

5 Results and what they mean

Exercise 1, Part (i) (from the report): American put price **0.11443**, European put price **0.05638** on the same 8 paths. The fitted regressions match the paper: at $t = 2$, $E[Y|X] = -1.06999 + 2.98341X - 1.81358X^2$; at $t = 1$, $E[Y|X] = 2.03751 - 3.33544X + 1.35646X^2$. Stopping rule and option cash-flow matrix reproduce the paper's table.

Exercise 1, Part (ii) (report Fig. 13): the 20-row replication of the red-box columns; e.g. $S = 36, sigma = 0.2, T = 1$ gives Simulated American **4.4714 (s.e. 0.0094)**, closed-form European **3.8443**, early-exercise value **0.6271**. s.e. all in the 0.006–0.023 range.

Exercise 2, Part (i) (report Fig. 14): FD-European vs BS-European agree to ~1e-4 across all 20 rows (largest $|Difference| = 0.0008$ at $S = 40, sigma = 0.2, T = 1$).

Exercise 2, Part (ii) (report Fig. 16): FD-American prices, all above the corresponding European, with early-exercise values matching the LSM ones in sign and magnitude.

Exercise 2 cross-check (report Fig. 17): $|FDAmerican - LSMAmerican|$ is below 0.05 for all 20 rows (max ~0.016), and the Part (iii) $FDEE - LSMEE$ differences are small and alternate in sign.

TODO (Bruno): The FD-European prices are *systematically slightly below* BS (every $Difference$ is negative, ~ -1e-4 to -8e-4). Is that a bias of the explicit scheme, the truncated domain, or the interpolation? Pick one and defend it — Fang will ask why the sign is consistent.

TODO (Bruno): The FD-American vs LSM-American differences are ~0.005–0.016, an order of magnitude larger than the FD-European vs BS error (~1e-4). Why is the American comparison so much looser? (Hint: LSM bias + 50-date Bermudan + regression noise vs a near-exact PDE.)

---

6 The hardest question Fang could ask here — and my answer

Likely hardest: "Your FTCS scheme is explicit. Is it stable for $Nx = 1000$**,** $Nt = 40000 * T$**? Derive the condition and check it for your worst-case** $sigma = 0.4$**."**

Factual setup from the code: in log-space the diffusion coefficient is $0.5 * sigma^2$, `h = L/Nx` with $L = 2logK + 2 = 2log40 + 2 \approx 9.376$, so $h \approx 0.00938$; `k = T/Nt = 1/40000 = 2.5e-5` per unit time.

The explicit-scheme stability constraint for the diffusion part is roughly `0.5*sigma^2 * k / h^2 <= 1/2`, i.e. `k <= h^2 / sigma^2`.

TODO (Bruno): Plug in the numbers: `h^2 / sigma^2 = (0.00938)^2 / 0.16` $\approx$ `5.5e-4`, and $k = 2.5e - 5 << 5.5e - 4$, so the diffusion number $\approx 0.045 << 0.5$. State this out loud, confirm the scheme is comfortably stable, and explain *why the author picked Nt so large* (was it stability, or matching the paper's accuracy?). Also be ready to say what the $(1 - rk)$ reaction term and the $T1$ advection term contribute to the stability bound — and whether the advection (first-derivative) term can cause spurious oscillations even when the diffusion bound is met.

Backup hardest (LSM): "Is your LSM price biased high or low, and why?" TODO (Bruno): Standard result — using the *same* paths to estimate the continuation regression and to value the option introduces a look-ahead/in-sample bias in the continuation estimate, so the exercise decision uses noisy information. State the direction of the resulting price bias and how an out-of-sample / two-pass scheme would fix it.

---

7 "Show me in your code where you do X" — anticipated spots

- **"... filter to in-the-money paths before regressing."** `exercise1_def (2).py:102-104` ($itm_indexes = S[:, i+1] < K$), and `:211-213` for Part (ii).
- **"... build the discounted continuation target Y."** `:99-101` (Part i) and `:207-209` / `:182-184` (Part ii).
- **"... fit the continuation-value regression and read off the coefficients."** `:107-112` (monomial pipeline) and `:215-217` (Laguerre).
- **"... make the exercise-vs-continue decision and zero out future cash flows."** `:122-127` (Part i), `:224-229` / `:194-199` (Part ii).
- **"... compute the American and European prices."** `:147-157` (Part i American + European).
- **"... generate antithetic paths."** `:177-178`.
- **"... evaluate the weighted Laguerre basis."** `:185-191`.
- **"... the closed-form Black–Scholes put."** `:193-196` (Ex.1), `exercise2_def (2).py:24-26` and `:74-76`.
- **"... assemble the explicit FTCS evolution matrix F."** `exercise2_def (2).py:35-40` (and `:85-90`).
- **"... set the initial/boundary conditions of the PDE."** initial: `:45-46`; boundary term: `:51` (and `:103`).
- **"... the one line that makes it American."** `exercise2_def (2).py:105` (`U = np.maximum(U, intrinsic)`).
- **"... read the price off the grid at S0."** `:54` and `:107` (`np.interp(np.log(S0), mvec, U[:, Nt])`).
- **"... cross-check FD against LSM."** `:235-248` ($|FDAmerican - LSMAmerican|$).

---

8 Corrector feedback (from the graded PDF) — points lost and what to do

Extracted from the grader's annotations in `Computational_Finance_Set_2-3262965.pdf`. Two items CONFIRM things this sheet only raised as TODO/uncertain — they are settled now.

Ex. 1 — LSM - −0.2 (formatting): *"The format is not very good... in the table 'Cash-flow matrix at time 2', please keep all numbers to 5 digits after the decimal point, not only half numbers."* - −**0.5 — `np.abs` on the Early-Exercise Value was WRONG (CONFIRMS this sheet's TODO):** *"For the Early Exercise Value, you do not need to use the absolute value (`np.abs`)."* → My Q on `:257`/`:223` (abs vs signed) is now settled: the **signed** difference is correct; remove `np.abs`. The early-exercise value should be *American* − *European* (and it can legitimately be reported as-is; absolute value is unnecessary). - −**0.5 — efficiency:** *"the code is not very efficient, try to vectorize some for loops."* - −**1.0 — antithetic standard error was WRONG (CONFIRMS this sheet's TODO):** *"The s.e. calculation is incorrect: for antithetic Monte Carlo, the standard error should be computed from the 50,000 pair-averaged payoffs, not from the 100,000 individual path payoffs. I think the standard error in the paper is also incorrect, and some other literature has also mentioned this error in this paper."* → My `:258` s.e. TODO is settled: `std(cashflow)/√N` over the **100,000** individual paths is **wrong** because antithetic pairs are not independent. Correct s.e.: average each antithetic **pair** into one payoff (→ 50,000 i.i.d. pair-averages), then `std(pair_means)/√50000`. **Bonus point to make out loud:** the grader agrees the paper's own s.e. is also incorrect.

Ex. 2 — Finite differences - −1.0 — FD efficiency: *"The FD matrix F is a tridiagonal matrix, you do not need to store the full matrix and apply the full matrix-vector multiplication at each time step."* → Use a tridiagonal (banded) representation / sparse solve, not a dense $Nx \times Nx$ matrix. - −**1.5 — the FD-vs-LSM "test":** *"LSM is a kind of Monte Carlo, it has sampling error, so it is not a deterministic benchmark. And it would be better to add some more tests, for example, if the American price is always larger than or equal to the European price."* → Don't present LSM as the ground truth; add the **American ≥ European** sanity check (and ideally compare FD against a finer FD / closed form). - −**1.0 — exercise-date assumption:** *"In the paper the authors assumed the American option can be exercised 50 times a year... but you assumed it can be exercised 40000 times a year, making your American option price slightly larger than in the paper."* → Your $Nt = 40000 \cdot T$ makes the FD effectively continuously-exercisable; to match the paper's Bermudan you should allow exercise at **50/year**. Be ready to explain why more exercise opportunities raise the American price.

Action checklist: (1) drop `np.abs` on EE (signed); (2) recompute the antithetic s.e. from 50,000 pair-averages (and mention the paper's error); (3) vectorize / use a tridiagonal FD solve; (4) stop calling LSM a deterministic benchmark, add American ≥ European; (5) reconcile 40000 vs 50 exercise dates.